

H-60 Phase 1 — Base-Class Consolidation: Detailed Plan

Prepared: 2026-08-17 · **Basis:** master improvement plan §Phase 1 (corrected Snow plan), the verified UH60↔uh60_config structural diff (2026-08-17), and Riverman's rulings on the Slick armament, MLASS lifetime, and HH60 separation. **Prerequisite:** Phase 0 merged (PRs #561–#566 + the two MP branches).

1. Objective

Today vtx_H60_base is **fully declared in four addons** (UH60, uh60_config, H60_SFX, ace_viv) and three variants (vtx_UH60M, vtx_UH60M_SLICK, vtx_HH60) are **declared twice** (UH60 + uh60_config). Whoever loads last silently wins every property collision. This is how the Slick's minigun visibility got inverted without anyone noticing, and it is the leading suspect for issue **#510** (cockpit sound attenuation loss).

Phase 1 ends with **one full declaration per class, everywhere** — the rule both reference projects (AH-64D, MH-47G) enforce without exception:

- Every other touchpoint becomes an honest **delta**: forward-declare the parent, re-open the class, add only your own members.
- Behavior after the change is **provably identical** to before (config-dump diff), except where a divergence is resolved deliberately and on the record.

Phase 1 does **not** merge addons, rename classes, move assets, or delete dead code (that's Phase 2). It is a declaration-ownership cleanup plus one structural move: the HH60's departure into its own addon.

2. Target end-state

Addon	Role after Phase 1
UH60	Assets (unchanged) + the only full declarations of vtx_H60_base and the Army family : vtx_UH60M, vtx_UH60M_SLICK
HH60 (new addon)	The Air Force bird : sole owner of vtx_HH60 (config-only; flies the shared p3d until the HH-60W model conversion)
uh60_config	Honest systems-delta layer: TextureSources (liveries), MASS_* weight logic, granular seat sources, hct_turret panels, FFV windows, CBA functions; sole owner of vtx_UH60M_MEDEVAC and vtx_S70M (unchanged — never duplicated)
H60_SFX	Sound-only delta on vtx_H60_base (its current full redeclaration demoted)
ace_viv	Unchanged (already a one-property delta; see §6 warnings)
MH60M / MH60S	Unchanged structurally; requiredAddons corrected
Other 11 re-openers (mfd, fms, anvishud, sfmplus, acre, cas, hoist, misc, weapons, ui, aar)	Unchanged — they are already well-behaved deltas

Variant identity rulings that govern the merge:

- **vtx_UH60M_SLICK is the armed ViV-cargo variant.** uh60_config's values (miniguns **shown**) are the intent; UH60's hide-lines are the wrong copy and are deleted. (Riverman, 2026-08-17)

- `vtx_UH60M_MEDEVAC` (HH-60M) is the only unarmed military variant. `vtx_S70M` is civilian, unarmed.
 - `vtx_HH60` is currently a G-dressed airframe with a W-style cockpit; its long-term identity is the HH-60W. No G-specific fidelity work; the class name does not change in Phase 1 (rename decision deferred to the model conversion — consider a scope-1 compat alias then).
 - **MLASS stays flyable** until its planned removal; Phase 1 must not degrade it.
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3. Work packages, in order

WP0 — Verification harness (build first, use throughout)

1. **Config-dump script:** for each spawnable class, dump every property with **values** (the master plan's snippet — `configProperties` with `getNumber/getText/getArray`; the Snow plan's names-only version misses silent value changes).
2. **Baseline capture on Main** for the full spawn matrix: `vtx_UH60M`, `vtx_UH60M_SLICK`, `vtx_UH60M_MEDEVAC`, `vtx_HH60`, `vtx_S70M` (*the Snow plan omitted it*), `vtx_MH60M`, `vtx_MH60M_DAP`, `vtx_MH60M_DAP_MLASS`, and the MH60S set.
3. Store dumps as files; the acceptance test for every WP below is `diff old new` = empty except the deliberate changes listed in that WP.

WP1 — `vtx_H60_base`: one owner

1. UH60's declaration ([UH60/config/cfgVehicles.hpp:38-458]) becomes the single source.
2. **H60_SFX** (`config.cpp:200-312`): demote to a delta carrying *only* its sound content (`soundEngineOn*`, `class sounds`, `SoundSets`). Two traps from the corrected plan:
3. `playerPosition = 1` (`config.cpp:197`) sits on the **vanilla** `Heli_Transport_01_base_F` → `CargoTurret` — it must **not** move into `vtx_H60_base`.
4. H60_SFX redeclares **76** `vtx_*` classes in total; sweep `Compat.hpp` siblings into delta form in the same pass.
5. **ace_viv**: stays exactly as-is (see §6 — do not "clean up" its GVAR macro).
6. **Base-level value collisions** — resolve to the later-loading (live) value, then delete the loser:
7. `RotorHFOld`: keep `animPeriod=5`; `initPhase=0` (`uh60_config`), delete UH60's `1 / -0.125`.
8. The ~60 duplicated `AnimationSources` between the two `AnimationSources` blocks: keep one copy of each; `uh60_config` retains only sources it genuinely adds or extends (`mass=`, `forceAnimate`, `onPhaseChanged`, `blade-fold set`, `windows`, `wipers`).
9. `uh60_config`'s `vtx_H60_base` delta keeps its legitimate content: `class Armory` blank, `TextureSources` (12 liveries), `hct_turret_1/2/3`, its class EHs and CBA functions. (*The corrected plan explicitly rejects making it a bare stub — delta-extension from a later loader is the reference-blessed pattern.*)

Deliberate changes to log: none expected at runtime — WP1 should be config-dump **byte-identical** for every variant.

WP2 — Army variant merges (`vtx_UH60M`, `vtx_UH60M_SLICK`)

1. **`vtx_UH60M`:** the two halves are disjoint (identity/visual in UH60; seats/MFD/medical in `uh60_config`). Keep the full declaration in UH60; `uh60_config`'s block remains as the delta that wires MFD elements (it must stay in `uh60_config` — it references `uh60_mfd/fms/anvishud` content that loads after UH60). Net config: identical.
2. **`vtx_UH60M_SLICK`:** merge per the armament ruling — miniguns/mounts/gunner seats **shown**. Delete UH60's five `ANIM_INIT(...,1)` hide lines (UH60:564-568); keep `uh60_config`'s `initPhase=0` set and granular `CabinSeats_1/2/3`. Remove the now-dead master `CabinSeats_Hide,1` and `Hoist_hide/sight` lines only if the config dump proves them inert; otherwise keep and log.

3. **Audit the Slick's turret-lock indexes:** `uh60_config's cabindoor_L/R onPhaseChanged` locks turrets `[[3], [4]]/[[1],[2]]`, which matches neither HH60 (`[[5],[6]]`) nor MEDEVAC mappings. Verify against the merged Slick's actual Turrets order (UH60's block contributes `cargoTurretsDoor.hpp` — 7 door cargo turrets) and correct. **This is a deliberate change if wrong today; test door-lock behavior per seat in-game.**
4. **MEDEVAC placement decision** (*small, Riverman's call during review*): `vtx_UH60M_MEDEVAC` can stay wholly in `uh60_config` (it is not duplicated; least churn — recommended) or move to UH60 beside its siblings for family symmetry. Recommendation: **stay**, revisit in Phase 4.

Deliberate changes to log: Slick turret-lock indexes (if corrected). Everything else byte-identical.

WP3 — HH60 separation (new addon)

1. Create addons/HH60 (`CfgPatches vtx_HH60_addon` or similar — avoid colliding with the vehicle class name), `requiredAddons[] = {"vtx_UH60", "vtx_UH60_mfd", "vtx_UH60_fms", "vtx_UH60_anvishud"}` — the same set `uh60_config` needs, because the HH60 is the only variant wiring the full-FLIR `VTX_MFD_1..4`.
2. Move **both halves** into it: UH60's full declaration (UH60:498-550) + `uh60_config's delta` (`uh60_config:340-401`), merged into one clean declaration deriving from `vtx_H60_base` — the MH60M/MH60S pattern.
3. Resolve its one conflict on the record: `CabinSeats_Hide` → `initPhase=0` (`uh60_config's` granular replacement is the live behavior; UH60's `,1` line was dead-but-misleading).
4. `units[]`: remove `vtx_HH60` from UH60's `CfgPatches`, add to the new addon's.
5. **Assets stay in UH60** (shared `vtx_uh60.p3d`, `pavehawk.hpp` model.cfg fragment, HH60/HH60_Fire textures) — path references need no dependency edge. The new addon is config-only until the HH-60W model work.
6. **Load-order fix for patchers of vtx_HH60:** `H60_SFX` (`Compat.hpp` adds `CargoTurret_01` to `vtx_HH60`) currently requires only `vtx_uh60+vtx_uh60_config`. Either add the new HH60 addon to `H60_SFX's` `requiredAddons`, or move that one compat block into the new addon. Recommendation: add the dependency — keeps sound content in the sound addon.
7. `uh60_config` keeps **nothing** HH60-specific afterward.

Deliberate changes to log: none at runtime — the merged HH60 config must dump identically to today's post-load-order result.

WP4 — requiredAddons and inherit-header cleanup

1. Add `"vtx_UH60"` to `MH60M/config.cpp:9` and `MH60S/config.cpp:9` (they derive from `vtx_H60_base` but never declare the dependency — works today only by accident of load order).
2. `CfgAnimationSourcesInherit.hpp` (UH60; consumed only by MH60M/MH60S): with single ownership established this forward-declaration crutch may be simplifiable — **investigate, but defer any change to Phase 2/4 if it isn't a pure win**. It is not required for Phase 1's goal.
3. While in the area, note (do not fix) Phase 2 candidates uncovered by the diff: the disabled parallel livery system (`vehicleCustomization.hpp`, include commented at UH60:87), `units[]` case mismatch `vtx_UH60M_Slick` vs `SLICK` (harmless), dead `doorguns.hpp`.

4. Branch & PR strategy

The master plan scoped Phase 1 as one PR; with the HH60 amendment, three **sequential** PRs keep each diff reviewable and independently regression-tested, at the cost of the middle state shipping nothing user-visible:

1. `restructure/base-single-owner` — WP1 (base class + `H60_SFX` demotion). Config-dump proof: byte-identical.
2. `restructure/army-variant-merge` — WP2. Proof: byte-identical except logged Slick items.

3. `restructure/hh60-addon` — WP3 + WP4. Proof: byte-identical per variant; spawn matrix green; #510 retest.

Same workflow as Phase 0: branch published at creation, commits local until Riverman verifies, PR on his word. Each PR's build = full `scons a11` (25 PBOs, 26 after the HH60 addon lands) plus the config-dump diff attached to the PR description.

5. Verification & acceptance

- **Config-dump diff** (WP0 harness) — the primary proof, run per-variant per-PR.
 - **Spawn matrix** — every class in WP0's list spawns in Eden/Zeus, textures and seats correct, no RPT errors. Include getting in pilot/copilot/door-gun/cargo seats.
 - **Issue #510 as acceptance test** (*from the master plan*): cockpit sound attenuation depends on turret-config merges across the four competing base declarations; if Phase 1 is done right, attenuation works (or its true cause is exposed). Test UH60M + MH60M cockpit audio before/after.
 - **Issue #439 retest** (hover symbology on UH/HH models) after WP3 — `vtx_HH60` was one of the duplicate-delta'd classes.
 - **Slick functional test**: door guns present and usable (armament ruling), ViV cargo loading works, door-lock behavior per seat (WP2.3).
 - **MLASS regression**: spawns, 4 pylons work, FLIR page carets/icons intact (Phase 0 #566 behavior preserved).
 - **MP smoke test** on a dedicated server for one Army variant + HH60: crew seats, doors, sounds — the load-order-sensitive surfaces.
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6. Do-NOT list (traps recorded in the corrected plan and diff)

1. **Do not move** `playerPosition = 1` (H60_SFX config.cpp:197) into `vtx_H60_base` — it belongs to the vanilla CargoTurret path.
 2. **Do not copy** `GVAR(lockSeats)` **text between addons** — in `ace_viv` it expands to `vtx_ace_viv_lockSeats`; pasted elsewhere it expands differently and silently breaks `ace_viv`'s scripts. Leave `ace_viv` untouched.
 3. **Do not re-open a populated nested container from a second addon** (the MH-47G "shadow-not-merge law"): re-opening `class Turrets` replaces the whole tree. Deltas add top-level members or extend via forward-declared inner classes only.
 4. **Do not resolve collisions toward UH60's values** — the later loader's value is today's live behavior; keeping UH60's would change the aircraft (the Slick incident is the proof).
 5. **Do not relocate the MASS_* macros casually** — they live in `uh60_config/script_component.hpp:20-42` and feed its `AnimationSources`; if any animation-source file moves addons, the defines must move to a shared header first. `hct_minigun.hpp` additionally depends on `POS_MOVE_CABIN_COCKPIT` (`main/script_macros.hpp`) — verify include chains before moving it.
 6. **Do not touch MLASS-specific content beyond keeping it working** — it is scheduled for removal; don't invest, don't break.
 7. **Do not rename** `vtx_HH60` in this phase — mission/save compatibility; the rename rides with the HH-60W model conversion.
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7. Risks & mitigations

Risk	Mitigation
Silent value change survives the merge	Value-capturing config dump (not names-only); per-variant diffs attached to PRs
Load-order patcher breaks when vtx_HH60 moves (H60_SFX Compat, missions referencing the class)	Class name unchanged; explicit requiredAddons edge; spawn matrix + #439/#510 retests
Slick turret-lock indexes were wrong all along	Treated as deliberate, tested change (WP2.3), not silently "fixed"
H60_SFX demotion drops a sound property	Its 76-class redeclaration sweep is diffed class-by-class; #510 test doubles as the audio acceptance gate
Regression discovered post-merge	Three small sequential PRs — each independently revertible; bisection stays cheap

8. Open items for Riverman

1. Name for the new addon folder/CfgPatches class (addons/HH60, patches class distinct from the vehicle class).
2. MEDEVAC declaration placement (recommend: stays in uh60_config for Phase 1).
3. Confirm the three-PR split vs the original single-PR scope.
4. When WP2.3's turret-lock audit produces its answer: confirm the intended door-lock behavior per Slick seat.